ZHANHONG HE

Music and Audio Machine Learning Researcher

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https://github.com/zhanh-he/ | https://www.linkedin.com/in/zhanhong-he/

Research Profile

Music and audio ML researcher studying how audio and symbolic representations capture musical expression. Research covers cross-instrument MIDI velocity estimation, score-informed transcription refinement and generative modelling, with professional listening-test experience. Interested in musician-facing tools for performance, analysis and composition.

Selected Research

Cross-instrument musical dynamics

- Contributed to a differentiable SoundFont proxy approach for adapting a piano-pretrained velocity estimator to other instruments using audio-loudness supervision.
- Investigated expressive rendering where reliable non-piano velocity labels are scarce.

Score-informed transcription refinement

- Contributed to a Transformer correction approach that uses score onset and frame information to refine MIDI velocities produced by piano transcription models.

Joint dynamics and musical structure

- Contributed to a multitask CRNN for dynamics, beats, downbeats and structural changes, and to reimplementing a Bark-scale specific-loudness extractor in PyTorch.

MIDI expression generation and evaluation

- Contributed to adapting an image-colourisation U-Net to fill per-note MIDI velocities; designed a listening study conducted with support from Dolby professional listeners.

Project repository: https://github.com/zhanh-he/sfproxy-velocity-estimation

Research Experience

Huawei Hongkong Research Centre | Research Intern, full time

Appointment commencing June 2026; six-month appointment.

- Conducted research on music-model post-training and human-preference evaluation.

Dolby Laboratories, Sydney | Research Intern, full time | May-September 2024

- Developed music-generation models using U-Net, VAE and masked Transformer architectures.
- Designed and conducted professional listening tests for musical audio.

Great Bay University | Visiting Student Researcher, part time

Appointment commencing March 2026; three-month appointment.

- Explored pretrained music representations, including MERT, and agentic workflows for MIR research.

Education

University of Western Australia | PhD research in Electrical and Electronic Engineering

February 2022-present. Doctoral defence completed; graduation certificate expected November 2026.

Research on musical dynamics, audio-symbolic modelling and machine learning. University International Fee Scholarship; University Postgraduate Awards; 2026 UWA Data Institute Award.

University of Western Australia | BSc Honours in Engineering Science

October 2018-December 2021. First Class Honours; GPA 7/7; WAM 87.7/100.

Honours thesis: Improving Spectrogram-based Music Source Separation using Phase Information.

Selected Publications

Zhanhong He, Hanyu Meng, David Defeng Huang and Roberto Togneri. Beyond Piano: Cross-Instrument MIDI Velocity Estimation via Differentiable SoundFont Proxies. Accepted to ISMIR 2026; arXiv:2608.08985. https://arxiv.org/abs/2608.08985

Zhanhong He, Hanyu Meng, David Huang and Roberto Togneri. Joint Estimation of Piano Dynamics and Metrical Structure with a Multi-task Multi-Scale Network. ICASSP, 2026. DOI: 10.1109/ICASSP55912.2026.11460839.

Zhanhong He, David Cooper, Defeng Huang and Roberto Togneri. Filling MIDI Velocity using U-Net Image Colorizer. Proceedings of CMMR 2025, pp. 949-960. https://arxiv.org/abs/2508.07751

Teaching And Music Practice

University of Western Australia | Tutor and Lab Facilitator | February 2023-December 2025

- Supported Signal Processing and Engineering Practices teaching; co-supervised Computer Science Honours projects and final-year designs.
- Guitar and piano enthusiast; UWA music coursework included electronic composition, Max/MSP music generation and ensemble performance.
- Built an Arduino-based MIDI theremin programmed in C at a Physical Interaction Design for Music workshop in 2023.

Technical Skills

Python; PyTorch; audio and symbolic music processing; MIDI; CRNN, Transformer, U-Net and VAE research; perceptual evaluation; Ableton Live; Max/MSP; basic C and C++.

Languages: English fluent; Chinese and Cantonese native.